

TECHNICAL ASSESSMENT

Semantic Similarity Search System: Design Brief

For the role of NLP Engineer

ISSUED BY	Nexora AI, Language Systems Group
ROLE	NLP Engineer (Product Engineering)
ASSESSMENT TYPE	Take-home design and implementation exercise
DATE OF ISSUE	30 June 2026
TIME ALLOWED	72 hours from receipt

1. CONTEXT

This assessment is for an NLP Engineer role at Nexora AI. The team builds systems that search and retrieve text based on meaning. The task below is similar to the work the team does. It is used to check how well a candidate can design, build, and explain a working solution.

2. PROBLEM STATEMENT

Design and implement a system that takes a user query and returns the documents that are most relevant to it. The match should be based on meaning, not on exact keyword matching.

3. OBJECTIVE

Given a user query and a set of text documents, the system shall return the documents that are closest to the query in meaning. The results shall be ranked, with the most relevant document first. The comparison shall use vector representations of the text rather than exact word overlap.

4. FUNCTIONAL REQUIREMENTS

4.1 Text Preprocessing

The system shall accept a text query and apply the same cleaning steps, such as lowercasing, tokenising, and removing stopwords, to both the query and all documents.

4.2 Text Representation

The system shall represent words and documents as dense vectors using word embeddings such as Word2Vec (CBOW or Skip-Gram).

4.3 Unseen Words

The system shall handle words that were not seen during training. FastText may be used for this.

4.4 Similarity and Ranking

The system shall measure semantic similarity between the query and each document using cosine similarity, and shall return the documents in ranked order, most similar first.

4.5 Baseline Comparison

The system shall compare its results with a simple keyword-based method (TF-IDF) to show the benefit of meaning-based search.

4.6 Generalisation

The system shall not be tuned to a single dataset and shall work on other text collections.

4.7 Number of Results

The system shall let the user set how many documents are returned for each query.

5. INPUT AND OUTPUT SPECIFICATION

Input: A user query in plain text, and a set of text documents to search.
Output: A ranked list of the most relevant documents, with the closest match first.

6. EVALUATION CRITERIA

Ref	Criterion	Description
6.1	Retrieval Quality	The returned documents are relevant to the query in meaning, and the most relevant ones appear first.
6.2	Robustness	The system handles different queries, messy text, and words not seen during training.
6.3	Generalisation	The system works on documents and queries that were not used during development.
6.4	Clarity of Approach	The candidate clearly explains the method and the reasons for the main choices.
6.5	Code Quality	The code is clear, organised, and easy to run.

7. SUBMISSION REQUIREMENTS

Candidates shall submit the following:

- Source code
- Report
- README
- Output screenshots
- Short explanation video

The explanation video is required. It shall cover: (a) the overall approach and the main choices; (b) how the system works, from preprocessing to ranking; and (c) a short run through one query, showing how the results are produced. A submission without the video will be treated as incomplete.

8. TIMELINE

The assessment shall be completed and submitted within 72 hours from the time of issue. Late submissions will not be considered unless an extension is agreed in writing in advance.

Note to candidate: You may use any public text dataset and any common Python libraries (for example Scikit-Learn, Gensim, Word2Vec, and FastText), as long as you list them in your explanation. If anything is unclear, please ask before you start.